So we did learn Kerberos on Day 20, but we connected the authentication flow.

1. DNS locates the DC.
2. DC authenticates the user.
3. Kerberos issues tickets.
4. AD determines group memberships.
5. Authorization occurs.
6. Group Policy is applied.
7. The user receives their desktop.

Next question: Now that the user is logged in, how is the Windows environment actually built and managed? That leads to Windows Server roles. Not just AD.

Day 21 Goal: Understand what Windows Server actually does inside an enterprise. Windows Server is not AD. AD has only one role.

Windows Server Roles

Refresher: Windows Server is an OS designed to manage computers, users, applications, and network resources in an organization.

A server role in Windows Server is a specific function that a server is configured to perform on a network. Each role installs the services, features, and management tools needed for that particular job. Instead of every server doing everything, Windows Server allows admins to assign only the roles that are needed.
Examples of common server roles:

- AD Domain Services (AD DS) - authenticates users and manages domains
- DNS Server
- DHCP Server
- File and Storage Services
- Web Server (IIS) - hosts websites and web applications
- Print Server
- Remote Desktop Services
- Hyper-V

Microsoft separates functionality into roles to make Windows Server more secure, efficient, and easier to manage.

How These Roles Work Together

Don’t think of the roles listed above individually. Think like an enterprise architect.
(See next page).

```
Internet
      |
      ▼
Firewall
      |
      ▼
Windows Server
├── Active Directory
├── DNS
├── DHCP
├── File Services
├── Print Services
└── Remote Management

        |
        ▼

Employees
├── HR PC
├── IT PC
├── Finance PC
└── Sales PC
```

When Kevin logs into his computer...Which server roles are involved?

AD, DNS, DHCP, File Services after login

When Kevin opens a shared company folder...Which server roles are involved?

File Services, AD, DNS Server (by helping Kevin's computer locate the file server if it connects using the server's hostname)

When Kevin plugs a new laptop into the network…Which server roles are involved?

AD, DNS, DHCP, Remote Management (IT use)

Making a Career Connection

A Windows Server Administrator is responsible for installing, configuring, securing, monitoring, and maintaining Windows Server systems so that an organization's IT infrastructure runs reliably and securely.